

Statistics & Sampling



Sampling methods

- 1 A school wants to survey students about the cafeteria. Explain why surveying only students in the chess club would produce biased results, and suggest a better sampling method.
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Comparing distributions using shape, center, and spread

- 2 Two data sets have the same median but Data Set A has a much larger range than Data Set B. Explain what this tells you about the variability of each data set.
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Linear regression and correlation

- 3 A scatterplot of hours studied vs. exam score has a line of best fit $y = 6.5x + 52$. Interpret the slope and y -intercept in context.
 - 4 A data set has a correlation coefficient of $r = -0.85$. Describe the strength and direction of the relationship.
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Using a line of best fit to make predictions

- 5 A line of best fit for predicting weight (in kg) from height (in cm) is $W = 0.9H - 70$. Predict the weight of a person who is 175 cm tall, and explain a limitation of using this model for a person who is 250 cm tall.
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Interpreting the standard deviation

- 6 Data Set A has standard deviation 2, and Data Set B has standard deviation 8, both with the same mean. Explain what this indicates about the two data sets, and give a real-world example of each.
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Margin of error and confidence in survey results

- 7 A survey estimates that 42% of a population favors a new policy, with a margin of error of 3%. A second survey with a larger sample size estimates the same statistic with a margin of error of 1.5%. Explain why the second survey's margin of error is smaller, and what this means for the precision of the estimate.
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Evaluating experimental design

- 8 A researcher wants to test whether a new fertilizer increases crop yield. Explain why using a control group (plants without the fertilizer) is essential to this experiment's design.
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Calculating and comparing means from grouped data

- 9 A class of 25 students has a mean test score of 78. A second class of 15 students has a mean score of 88. Find the combined mean score of all 40 students.
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Recognizing bias in survey questions and methods

- 10 A survey asks, "Don't you agree that the excellent new park should be funded?" Explain why this question is biased, and rewrite it in a neutral form.
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Comparing two data sets using boxplot-style summaries

- 11 Data Set A has minimum 10, $Q1 = 20$, median 30, $Q3 = 40$, maximum 50. Data Set B has minimum 15, $Q1 = 28$, median 30, $Q3 = 32$, maximum 45. Both sets have the same median. Compare their spread using the interquartile range (IQR).
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Extrapolation vs. interpolation

- 12 A line of best fit is built from data on shoe size (ranging from 6 to 12) and height. Explain the difference between using this model to predict height for a shoe size of 9 versus a shoe size of 20, referencing interpolation and extrapolation.
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Relative frequency and probability from a frequency table

- 13 A frequency table of exam grades shows: A: 8 students, B: 15 students, C: 12 students, D: 5 students. Find the relative frequency (as a percent) of students who earned a B or higher.
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Random assignment vs. random sampling

- 14 Explain the difference between random sampling and random assignment, and state which one allows a study to make causal claims.

A multi-part regression and residual analysis problem

- 15** This question has three parts. A line of best fit for predicting sales revenue (in thousands of dollars) from advertising spend (in thousands of dollars) is $R = 4.2A + 18$. (a) Predict the revenue for an advertising spend of 10 thousand dollars. (b) If the actual observed revenue at $A = 10$ was 55 thousand dollars, find the residual (actual minus predicted). (c) Explain what a positive residual indicates about this particular data point relative to the line of best fit.

Multiple choice — easy

- 16** A homeroom class of 27 students has a mean test score of 86. A second section of 38 students has a mean score of 96. Find the combined mean score of all 65 students, to the nearest hundredth.
- a. 91.00
 - b. 89.85
 - c. 93.85
 - d. 91.85
- 17** A chemistry class of 36 students has a mean test score of 92. A second section of 26 students has a mean score of 62. Find the combined mean score of all 62 students, to the nearest hundredth.
- a. 77.00
 - b. 79.42
 - c. 81.42
 - d. 77.42

- 18** A homeroom class of 40 students has a mean test score of 93. A second section of 26 students has a mean score of 91. Find the combined mean score of all 66 students, to the nearest hundredth.
- a. 90.21
 - b. 92.00
 - c. 92.21
 - d. 94.21
- 19** A history class of 15 students has a mean test score of 67. A second section of 21 students has a mean score of 73. Find the combined mean score of all 36 students, to the nearest hundredth.
- a. 70.00
 - b. 68.50
 - c. 72.50
 - d. 70.50
- 20** A history class of 20 students has a mean test score of 86. A second section of 29 students has a mean score of 96. Find the combined mean score of all 49 students, to the nearest hundredth.
- a. 89.92
 - b. 93.92
 - c. 91.00
 - d. 91.92
- 21** An algebra class of 32 students has a mean test score of 98. A second section of 13 students has a mean score of 71. Find the combined mean score of all 45 students, to the nearest hundredth.
- a. 92.20
 - b. 84.50
 - c. 90.20
 - d. 88.20

- 22** A biology class of 27 students has a mean test score of 95. A second section of 13 students has a mean score of 88. Find the combined mean score of all 40 students, to the nearest hundredth.
- a. 92.72
 - b. 94.72
 - c. 91.50
 - d. 90.72
- 23** A chemistry class of 27 students has a mean test score of 75. A second section of 10 students has a mean score of 60. Find the combined mean score of all 37 students, to the nearest hundredth.
- a. 67.50
 - b. 72.95
 - c. 68.95
 - d. 70.95
- 24** An algebra class of 30 students has a mean test score of 81. A second section of 18 students has a mean score of 98. Find the combined mean score of all 48 students, to the nearest hundredth.
- a. 89.38
 - b. 89.50
 - c. 85.38
 - d. 87.38
- 25** A history class of 29 students has a mean test score of 92. A second section of 10 students has a mean score of 67. Find the combined mean score of all 39 students, to the nearest hundredth.
- a. 85.59
 - b. 83.59
 - c. 87.59
 - d. 79.50

- 26 An algebra class of 32 students has a mean test score of 83. A second section of 25 students has a mean score of 91. Find the combined mean score of all 57 students, to the nearest hundredth.
- a. 88.51
 - b. 84.51
 - c. 87.00
 - d. 86.51
- 27 A homeroom class of 12 students has a mean test score of 62. A second section of 14 students has a mean score of 78. Find the combined mean score of all 26 students, to the nearest hundredth.
- a. 70.00
 - b. 72.62
 - c. 68.62
 - d. 70.62
- 28 A homeroom class of 35 students has a mean test score of 76. A second section of 26 students has a mean score of 60. Find the combined mean score of all 61 students, to the nearest hundredth.
- a. 68.00
 - b. 67.18
 - c. 71.18
 - d. 69.18
- 29 A homeroom class of 17 students has a mean test score of 63. A second section of 29 students has a mean score of 74. Find the combined mean score of all 46 students, to the nearest hundredth.
- a. 68.50
 - b. 67.93
 - c. 71.93
 - d. 69.93

- 30 An algebra class of 10 students has a mean test score of 82. A second section of 10 students has a mean score of 73. Find the combined mean score of all 20 students, to the nearest hundredth.
- a. 78.50
 - b. 75.50
 - c. 77.50
 - d. 79.50
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Multiple choice — medium

- 31 A line of best fit for predicting race time (minutes) from training hours is $y = -0.50x - 15$. Predict the value of y when $x = 5$.
- a. -17.50
 - b. -18.00
 - c. -19.50
 - d. -15.50
- 32 A line of best fit for predicting sales revenue (thousands of dollars) from advertising spend (thousands of dollars) is $y = 4.00x + 62$. Predict the value of y when $x = 18$.
- a. 134.00
 - b. 132.00
 - c. 138.00
 - d. 136.00

- 33 A line of best fit for predicting exam score from hours studied is $y = -0.50x + 41$. Predict the value of y when $x = 2$.
- a. 42.00
 - b. 39.50
 - c. 40.00
 - d. 38.00
- 34 A line of best fit for predicting ice cream sales (dollars) from temperature (degrees F) is $y = -4.50x + 50$. Predict the value of y when $x = 18$.
- a. -35.50
 - b. -29.00
 - c. -33.00
 - d. -31.00
- 35 A line of best fit for predicting salary (thousands of dollars) from years of experience is $y = -1.50x + 59$. Predict the value of y when $x = 20$.
- a. 27.00
 - b. 31.00
 - c. 27.50
 - d. 29.00
- 36 A line of best fit for predicting sales revenue (thousands of dollars) from advertising spend (thousands of dollars) is $y = 4.50x + 7$. Predict the value of y when $x = 18$.
- a. 92.50
 - b. 90.00
 - c. 86.00
 - d. 88.00

- 37 A line of best fit for predicting salary (thousands of dollars) from years of experience is $y = -1.00x - 26$. Predict the value of y when $x = 2$.
- a. -28.00
 - b. -29.00
 - c. -30.00
 - d. -26.00
- 38 A line of best fit for predicting salary (thousands of dollars) from years of experience is $y = -4.00x + 38$. Predict the value of y when $x = 11$.
- a. -4.00
 - b. -8.00
 - c. -10.00
 - d. -6.00
- 39 A line of best fit for predicting race time (minutes) from training hours is $y = -1.50x + 10$. Predict the value of y when $x = 18$.
- a. -17.00
 - b. -15.00
 - c. -18.50
 - d. -19.00
- 40 A line of best fit for predicting race time (minutes) from training hours is $y = 1.50x - 12$. Predict the value of y when $x = 12$.
- a. 4.00
 - b. 8.00
 - c. 6.00
 - d. 7.50

- 41 A line of best fit for predicting exam score from hours studied is $y = -2.00x + 72$. Predict the value of y when $x = 11$.
- a. 51.00
 - b. 48.00
 - c. 52.00
 - d. 50.00
- 42 A line of best fit for predicting sales revenue (thousands of dollars) from advertising spend (thousands of dollars) is $y = -2.50x + 31$. Predict the value of y when $x = 12$.
- a. -1.00
 - b. -1.50
 - c. 3.00
 - d. 1.00
- 43 A line of best fit for predicting ice cream sales (dollars) from temperature (degrees F) is $y = -3.50x + 22$. Predict the value of y when $x = 5$.
- a. 4.50
 - b. 1.00
 - c. 2.50
 - d. 6.50
- 44 A line of best fit for predicting exam score from hours studied is $y = -4.00x + 24$. Predict the value of y when $x = 6$.
- a. -2.00
 - b. 2.00
 - c. 0.00
 - d. -4.00

- 45 A line of best fit for predicting race time (minutes) from training hours is $y = -3.00x + 35$. Predict the value of y when $x = 4$.
- a. 23.00
 - b. 20.00
 - c. 21.00
 - d. 25.00
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Multiple choice — hard

- 46 A data set has minimum 9, $Q1 = 17$, median 25, $Q3 = 31$, maximum 44. Find the interquartile range (IQR).
- a. 14
 - b. 16
 - c. 35
 - d. 12
- 47 A data set has minimum 19, $Q1 = 23$, median 47, $Q3 = 55$, maximum 65. Find the interquartile range (IQR).
- a. 46
 - b. 34
 - c. 30
 - d. 32

- 48 A data set has minimum 24, $Q1 = 35$, median 52, $Q3 = 71$, maximum 74. Find the interquartile range (IQR).
- a. 34
 - b. 38
 - c. 50
 - d. 36
- 49 A data set has minimum 28, $Q1 = 38$, median 54, $Q3 = 75$, maximum 84. Find the interquartile range (IQR).
- a. 37
 - b. 35
 - c. 56
 - d. 39
- 50 A data set has minimum 6, $Q1 = 12$, median 17, $Q3 = 18$, maximum 23. Find the interquartile range (IQR).
- a. 4
 - b. 6
 - c. 17
 - d. 8
- 51 A line of best fit is $y = 4x + 36$. At $x = 9$, the actual observed value is $y = 64$. Find the residual (actual minus predicted).
- a. -10
 - b. -6
 - c. 72
 - d. -8

- 52 A line of best fit is $y = -8x + 5$. At $x = 6$, the actual observed value is $y = -47$. Find the residual (actual minus predicted).
- a. -4
 - b. -6
 - c. -2
 - d. -43
- 53 A line of best fit is $y = 6x + 7$. At $x = 17$, the actual observed value is $y = 114$. Find the residual (actual minus predicted).
- a. 109
 - b. 5
 - c. 3
 - d. 7
- 54 A line of best fit is $y = -5x + 8$. At $x = 12$, the actual observed value is $y = -49$. Find the residual (actual minus predicted).
- a. 3
 - b. 1
 - c. 5
 - d. -52
- 55 A line of best fit is $y = -2x + 13$. At $x = 11$, the actual observed value is $y = -13$. Find the residual (actual minus predicted).
- a. -6
 - b. -4
 - c. -9
 - d. -2

- 56 A data set has $Q1 = 11$ and $Q3 = 21$. Using the $1.5 \times IQR$ rule, find the upper fence for identifying outliers.
- a. 21
 - b. 38
 - c. 36
 - d. 34
- 57 A data set has $Q1 = 21$ and $Q3 = 49$. Using the $1.5 \times IQR$ rule, find the upper fence for identifying outliers.
- a. 89
 - b. 49
 - c. 91
 - d. 93
- 58 A data set has $Q1 = 22$ and $Q3 = 60$. Using the $1.5 \times IQR$ rule, find the upper fence for identifying outliers.
- a. 60
 - b. 119
 - c. 117
 - d. 115
- 59 A data set has $Q1 = 37$ and $Q3 = 43$. Using the $1.5 \times IQR$ rule, find the upper fence for identifying outliers.
- a. 54
 - b. 50
 - c. 43
 - d. 52

- 60 A data set has $Q1 = 40$ and $Q3 = 50$. Using the $1.5 \times IQR$ rule, find the upper fence for identifying outliers.
- a. 67
 - b. 65
 - c. 50
 - d. 63